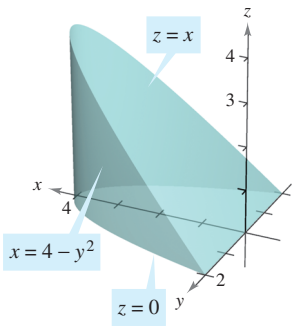
Suggested problems: 1-51 odd

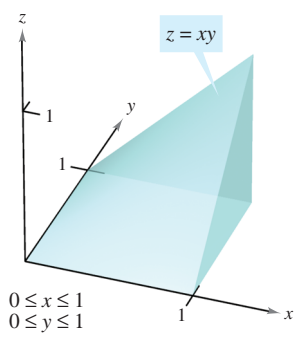
Problem 1: For each part below, set up the integral $\int_W f(x,y,z)\ dV$ where W is the region of xyz-space shown.

(a)



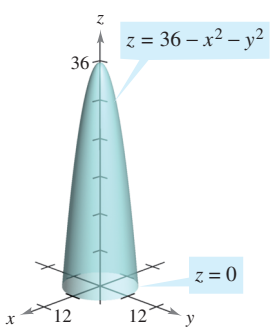
$$\int_{-2}^{2}\int_{0}^{4-y^2}\int_{0}^{x} f(x,y,z)\ dzdxdy$$

(b)



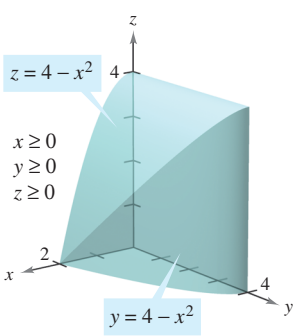
$$\int_{0}^{1}\int_{0}^{1}\int_{0}^{xy} f(x,y,z)\ dzdxdy$$

(c)



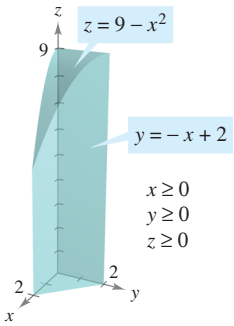
$$\int_{-6}^{6}\int_{-\sqrt{36-x^2}}^{\sqrt{36-x^2}}\int_{0}^{36-x^2-y^2} f(x,y,z)\ dzdydx$$

(d)



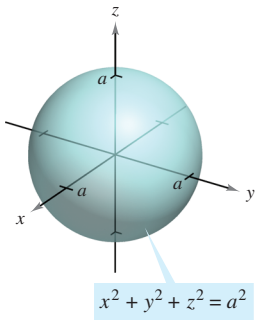
$$\int_{0}^{2}\int_{0}^{4-x^2}\int_{0}^{4-x^2} f(x,y,z)\ dzdydx$$

(e)



$$\int_0^2 \int_0^{-x+2} \int_0^{9-x^2} f(x, y, z)\ dzdydx$$

(f)



$$\int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_{-\sqrt{a^2-x^2-y^2}}^{\sqrt{a^2-x^2-y^2}} f(x, y, z)\ dzdydx$$

Problem 2: For each part below, sketch the region of integration.

(a) $\int_0^1 \int_{-1}^1 \int_0^{\sqrt{1-x^2}} f(x, y, z)\ dzdxdy$ See class notes.

(b) $\int_{-1}^1 \int_0^1 \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} f(x, y, z)\ dydzdx$ See class notes.

(c) $\int_0^1 \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} \int_0^{\sqrt{1-x^2-z^2}} f(x, y, z)\ dydxdz$ See class notes.